

Session 5 · Deck 5.B2

Choosing a Basis

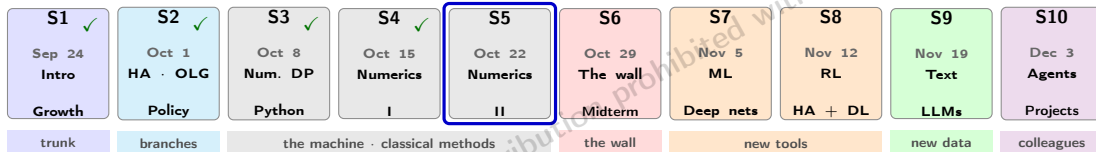
Zhigang Feng

Thursday, October 22, 2026

Build it on paper → solve it on the machine → push it to the frontier with AI

Where We Are

you are here: 5.B2 ▼



Build it on paper → solve it on the machine → push it to the frontier with AI

Block B: 5.B1 the residual · **5.B2** choosing a basis · 5.B3 Chebyshev · 5.B4 finite elements · 5.B5 determining the coefficients · 5.B6 the worked example.

Two Issues, and They Are Separable

1. Which basis $\{\psi_i\}$?

- A **global** basis — each ψ_i is non-zero and smooth almost everywhere on the domain $\Rightarrow$ **spectral methods**. Chebyshev polynomials, Fourier series.
- A **local** basis — each ψ_i is non-zero only on a small piece of the domain $\Rightarrow$ **finite element methods**. Tent functions, splines.

2. How do we “project”? Collocation, Galerkin, least squares, ... — 5.B5.

The two choices are independent, and the published methods are just particular pairs.

Quant_Macro Lec. 9, “Two issues” and “Spectral methods”. Main reference: Judd (1992), *JET* 58.

The trade

A global basis converges fast *if the function is smooth* and degrades badly if it is not. A local basis is indifferent to smoothness and converges slowly.

And they mix: spectral elements use a high-order basis *within* each element. So does a deep network, in effect.

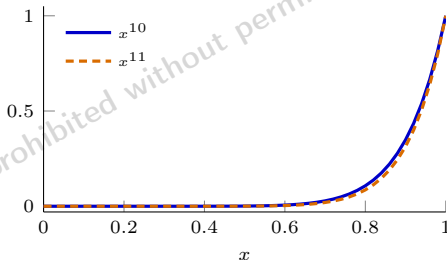
The Obvious Basis, and Why It Fails

Start with monomials: $1, x, x^2, x^3, \dots$

The theory is reassuring. By Stone–Weierstrass, every continuous function on a closed interval $[a, b]$ can be approximated *uniformly*, as closely as you like, by a polynomial. Nothing is missing from the family.

The practice is not. Compare the graphs of x^{10} and x^{11} on $[0, 1]$: they are nearly the same function. As basis directions they are 2.6° apart, so asking how much of each a fit needs is asking a question the data can barely answer.

And monomials differ enormously in **size** across the domain, so their coefficients do too, and rounding errors accumulate.



Approximation theory says the basis is fine. Numerical linear algebra says it is not. The second one wins.

Quant_Macro Lec. 9, "Polynomials". Ledger fig_mononials: under $\langle f, g \rangle = \int_0^1 fg \, dx$ the two are 2.605° apart.

Where That Matrix Comes From

Fit f on $[0, 1]$ with $p(x) = \sum_{j=0}^n \theta_j x^j$, choosing coefficients to minimise the squared error:

$$\min_{\theta} \int_0^1 \left(f(x) - \sum_j \theta_j x^j \right)^2 dx.$$

Differentiate with respect to θ_i , set to zero — one equation per coefficient:

$$\int_0^1 x^i \left(f(x) - \sum_j \theta_j x^j \right) dx = 0$$
$$\sum_j \theta_j \underbrace{\int_0^1 x^i x^j dx}_{H_{ij}} = \underbrace{\int_0^1 x^i f(x) dx}_{b_i}$$

That is $H\theta = b$: a linear system in the $n + 1$ unknown coefficients.

b depends on the function being fitted; H does not. Everything that follows is about H alone, so it is a property of the basis, not of the problem.

What H and b hold

H_{ij} is the **overlap of basis function i with basis function j** ; b_i is its overlap with the target. The system says: give the approximation the same overlap with every basis function that f has.

For monomials the integral is immediate:

$$H_{ij} = \int_0^1 x^{i+j} dx = \frac{1}{i+j+1},$$

which is the **Hilbert matrix** — the textbook example of an ill-conditioned system. Not an analogy: the same matrix.

What Goes Wrong: a Nearly Singular System

Nothing *visible* goes wrong: the solve succeeds and hands back $n + 1$ numbers. They are simply not pinned down.

Because x^{10} and x^{11} are almost the same function, raising θ_{10} and lowering θ_{11} by a similar amount leaves the fitted polynomial almost unchanged. So **very different coefficient vectors fit f about equally well**, and the fit cannot choose between them.

That is **collinear regressors**: fitted values fine, individual coefficients unidentified.

And the analogy holds all the way. At degree 13 the recovered coefficients are wrong by 180% while the fitted *polynomial* still matches to 2×10^{-8} . What is destroyed is θ — exactly what a projection method solves for and carries around.

H is a table of angles

Normalise an entry by its diagonal neighbours and you get the **cosine of the angle** between two basis functions:

$$\frac{H_{ij}}{\sqrt{H_{ii}H_{jj}}} = \cos \angle(x^i, x^j),$$

and they collapse together as the degree rises — which is why the trouble grows with n .

pair	cos	angle
x^1, x^2	0.9682	14.5°
x^5, x^6	0.9965	4.8°
x^{10}, x^{11}	0.9990	2.6°

Ledger basis_conditioning.hilbert_angles; angles under $\langle f, g \rangle = \int_0^1 fg \, dx$, the same inner product that built H .

Measured: How Much It Costs

$\text{cond}(H)$ is an **amplification factor**. Nudge the right-hand side and the solution can move by that much more:

$$\frac{\|\Delta\theta\|}{\|\theta\|} \leq \text{cond}(H) \frac{\|\Delta b\|}{\|b\|}.$$

You cannot avoid nudging it: merely *storing* b rounds it, by about one machine epsilon, 2.2×10^{-16} . So $\text{cond}(H) \times 2.2 \times 10^{-16}$ is the worst case for the answer, whatever solver you use.

It is a **worst case**, not a forecast — so we also recovered a known θ and measured what actually comes back.

Ledger basis_conditioning.hilbert and hilbert_recovery: a known θ , its exact $b = H\theta$, solved back in double precision. This is a property of the *basis*, not of the solver — no algorithm recovers information the matrix has destroyed.

n	$\text{cond}(H)$	bound on θ	measured
3	5.2×10^2	10^{-13}	10^{-16}
7	4.8×10^8	10^{-7}	5×10^{-9}
9	4.9×10^{11}	10^{-4}	2×10^{-6}
11	5.2×10^{14}	12%	0.6%
13	3.2×10^{18}	708	1.8

The bound runs about $100\times$ pessimistic, and the conclusion survives it: at degree 13 the recovered coefficients are wrong by **180%**. And **nothing fails** — no warning, no exception. The solver returns numbers.

Where the Other Matrix Comes From

Least squares was one way to pin down θ . The other is **interpolation**: pick $n + 1$ nodes and demand that the approximation match f *exactly* there,

$$p(x_i) = f(x_i), \quad i = 0, \dots, n.$$

Write p out in the basis and that is $n + 1$ linear equations:

$$\sum_{j=0}^n \theta_j \underbrace{\psi_j(x_i)}_{V_{ij}} = \underbrace{f(x_i)}_{f_i}$$

so $V\theta = f$, where V_{ij} is **basis function j evaluated at node i** . With monomials $\psi_j(x) = x^j$ that is $V_{ij} = x_i^j$ — the **Vandermonde matrix**.

Two matrices, one question

H was built from **integrals** — how much two basis functions overlap across the domain. V is built from **evaluations** — what each is worth at a point. Both end in a linear system for θ , and both inherit the basis's conditioning.

V hides two choices

Where you evaluate (the nodes) and **which** basis. They are separable, and it is worth knowing which does the work. V is also the matrix every collocation method in 5.B5–5.B6 forms.

Nodes below are the n Chebyshev points on $[-1, 1]$ in both columns — only the basis changes, so anything that differs is the basis's doing.

Measured: What an Orthogonal Basis Buys

Same experiment as before — recover a known θ — with the two choices varied one at a time. **Relative error in the recovered coefficients:**

n	monomial basis		Chebyshev
	equispaced	Cheb. nodes	Cheb. nodes
13	9×10^{-13}	4×10^{-13}	3×10^{-16}
21	4×10^{-9}	1×10^{-10}	7×10^{-16}
41	110%	1%	1×10^{-15}
61	280	150	2×10^{-15}

Better nodes buy an order of magnitude or two. The orthogonal basis buys the degree.

Below degree ~ 20 none of this matters — which is why monomials are survivable on a small problem. By degree 41 the monomial fit is worthless and the Chebyshev one has not moved. **It is what licenses the rest of the block:** 5.B3 expands to degree 33, 5.B6 solves at degree 11.

Why it never degrades

At the Chebyshev roots the columns of V are orthogonal, so $V^T V = \text{diag}(n, \frac{n}{2}, \dots, \frac{n}{2})$, measured to 2×10^{-14} off the diagonal. Hence $\text{cond}(V) = \sqrt{n/(n/2)} = \sqrt{2}$ at every n .

Ledger basis_conditioning.vandermonde_recovery and chebyshev_gram. $\text{cond}(V)$ at $n = 61$: 3.6×10^{18} equispaced, 1.9×10^{18} at Chebyshev nodes, 1.414 in the Chebyshev basis.

Why Orthogonality Helps, in Four Words Each

- **Simplicity.** Orthogonality means $H_{ij} = 0$ off the diagonal, so $H\theta = b$ **decouples**: the coordinates of a function *are* its projections onto the basis, and there is no system left to solve.
- **Independence.** Orthogonal functions are linearly independent by construction, so each one adds a genuinely new direction.
- **Numerical stability.** Small changes in the data produce small changes in the coefficients — which is exactly what the previous slide measured.
- **Interpretability.** Each coefficient means something on its own. This is the same property that makes principal components readable.

The formula

For $x = \sum_j c_j v_j$ with $\langle v_i, v_j \rangle = 0$ for $i \neq j$, take inner products with v_i : every term vanishes but one, leaving

$$c_i = \frac{\langle x, v_i \rangle}{\|v_i\|^2},$$

and simply $c_i = \langle x, v_i \rangle$ if the basis is orthonormal.

The Same Thing in $\mathbb{R}^2$

Take $x = (3, 4)$ and the orthonormal basis

$$v_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right), \quad v_2 = \left(\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right).$$

The coordinates are inner products, nothing more:

$$c_1 = \langle x, v_1 \rangle = \frac{3}{\sqrt{2}} + \frac{4}{\sqrt{2}} = \frac{7}{\sqrt{2}},$$

$$c_2 = \langle x, v_2 \rangle = -\frac{3}{\sqrt{2}} + \frac{4}{\sqrt{2}} = \frac{1}{\sqrt{2}},$$

so that $x = \frac{7}{\sqrt{2}}v_1 + \frac{1}{\sqrt{2}}v_2$.

In a non-orthogonal basis those two numbers would require solving a 2×2 system — and in a badly conditioned basis of degree 13, a system with no correct digits.

Quant_Macro Lec. 9, "Projections in an Orthogonal Basis" (worked example frame).

Function approximation is this, with $\langle f, g \rangle = \int fg w$ in place of the dot product. Nothing else changes.

Which is why Galerkin (5.B5) is natural: "make the residual orthogonal to every basis function" means "the residual has no component we could still represent".

Trigonometric Series

The other classical orthogonal family:

$$\frac{1}{\sqrt{2\pi}}, \frac{\cos x}{\sqrt{2\pi}}, \frac{\sin x}{\sqrt{2\pi}}, \dots, \frac{\cos kx}{\sqrt{2\pi}}, \frac{\sin kx}{\sqrt{2\pi}}, \dots$$

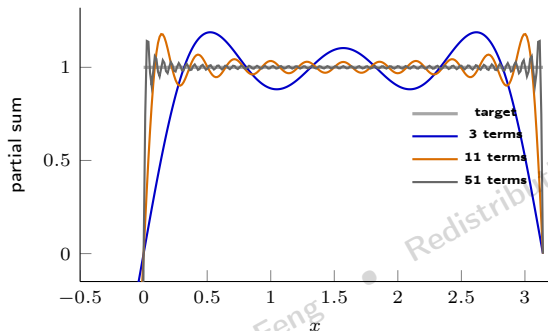
It is the right basis for **periodic** functions, and essentially

nothing in economics is periodic. A capital stock does not return to where it started after 2π .

Forcing a periodic basis onto a non-periodic function creates an artificial discontinuity at the ends of the interval — and with it the **Gibbs phenomenon**, which drags the convergence rate down to $O(1/n)$: the error falls like one over the number of terms, which is very slow.

Compare 5.B3: for a smooth function a Chebyshev basis converges *geometrically*. The gap between $1/n$ and r^{-n} is the whole argument.

The Gibbs Phenomenon, Measured



Fourier partial sums of a square wave. The overshoot beside the jump **does not shrink**:

terms	overshoot
3	18.84%
11	17.97%
51	17.90%
limit	17.898%

It converges to $\frac{2}{\pi}\text{Si}(\pi) - 1$ and stays there. More terms only move the oscillation *closer* to the jump.

Ledger basis_conditioning.gibbs. Measured against the half-amplitude; the textbook 8.95% is the same number measured against the full jump.

Why This Is a Macroeconomics Problem

Gibbs is usually filed under Fourier series, but the phenomenon is about **smooth global bases meeting non-smooth functions**, and polynomial bases have it too.

Macroeconomic policy and value functions are full of non-smoothness:

- **credit constraints** — a kink exactly at the borrowing limit (2.A1, 2.A2);
- **irreversible investment** — a kink at zero;
- **discrete choice** — an outright jump.

A Chebyshev expansion will overshoot near each of them, and **raising the order does not fix it**. It confines the oscillation to a smaller neighbourhood of the kink, at a larger amplitude nearby.

Quant_Macro Lec. 9, "Gibbs Phenomenon in Polynomial Approximation".

Two ways out

Piecewise bases — put an element boundary at the kink and the problem disappears. That is 5.B4. **Adaptive bases** — let the approximant learn where to bend. That is what a neural network does, and it is S7.

And it is why the Aiyagari economy is not solved with a global polynomial.

Takeaways

- Two independent choices: a **global** basis (spectral) or a **local** one (finite elements), and separately how to project.
- Monomials are complete by Stone–Weierstrass and **unusable** in practice: the least-squares matrix is the Hilbert matrix, with condition number 3×10^{18} at degree 13 — no correct digits, and no error message.
- An orthogonal basis fixes it completely. The Chebyshev interpolation matrix has condition number 1.414 at every degree tested, from 5 to 21.
- Orthogonality also makes coefficients into simple inner products, $c_i = \langle x, v_i \rangle / \|v_i\|^2$ — no system to solve, and each coefficient independently meaningful.
- Trigonometric bases suit periodic problems; economics has none. Forcing them gives $O(1/n)$ convergence, not the geometric rate a smooth problem deserves.
- **Gibbs does not go away with more terms**: measured at 18.8% with three terms and 17.90% with fifty-one, converging to 17.898%. Kinks need piecewise or adaptive bases, not higher order.

Bridge: The Basis That Wins

We now have the requirements: orthogonal, well conditioned at any degree, defined on an interval rather than a circle, and fast for smooth functions.

One family satisfies all of them, and it is the default in this literature for good measured reasons — including a theorem saying you cannot do much better with *any* polynomial of the same degree.

Next (5.B3)

Chebyshev polynomials: the recursion, the nodes, the interpolation theorem, what smoothness buys — and how the basis is extended to many dimensions, where Smolyak returns from 4.B1 as a *basis* rather than a node set.